

GMTK GAME JAM 2025

Hosted by Game Maker's Toolkit • #GMTKJam

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This jam is now over. It ran from July 30th 2025 at 7:00 PM to August 3rd 2025 at 8:00 PM. [View results](#)

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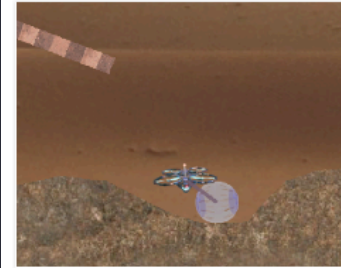
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Looking for [Godot 3](#), our [experimental releases](#), or a [previous version](#)?

<https://godotengine.org/>

Godot submissions

- Out of control, 2020
- Joined together, 2021
- Roll of the dice, 2022
- Role reversal, 2023
- Built to Scale, 2024



Unnamed Drone Game
Platformer
[Play in browser](#)



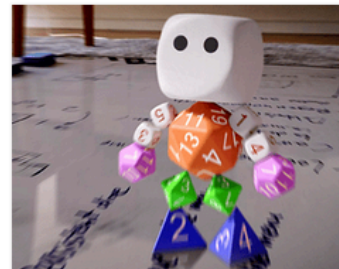
Super Scale Boy
Help Super Scale Boy get all the apples!
Platformer
[Play in browser](#)



Gunswinger
On-rail shooter where you aim by swinging your gondola left to right
Action
[Play in browser](#)



Tower offense
Run through your own gauntlet
Strategy
[Play in browser](#)



Henk and the legendary d20
Help Henk on his epic quest to find the legendary d20
Puzzle
[Play in browser](#)

Issues with Godot

- GDScript (~Python) makes me not want to write code
- 2020: Fiddling with Node2D physics tree, chain of joints unstable
- 2024: Theme=built to-scale, Godot doesn't support dynamic scaling.
- Bloat (40mb empty game)



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Let's use Rust! 🧑‍🚀 😊 😈

Dec 22, 2023 · 6 min read

5 Rust game engines to consider for your next project



Victor Jonah

I am a Software Developer with over three years of experience working with JavaScript and its frameworks. I currently work as a remote software developer for a tech agency.

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Why use Rust for game development?

Popular Rust game engines

Bevy

Fyrox



See how LogRocket's Galileo AI surfaces the most severe issues for you
No signup required

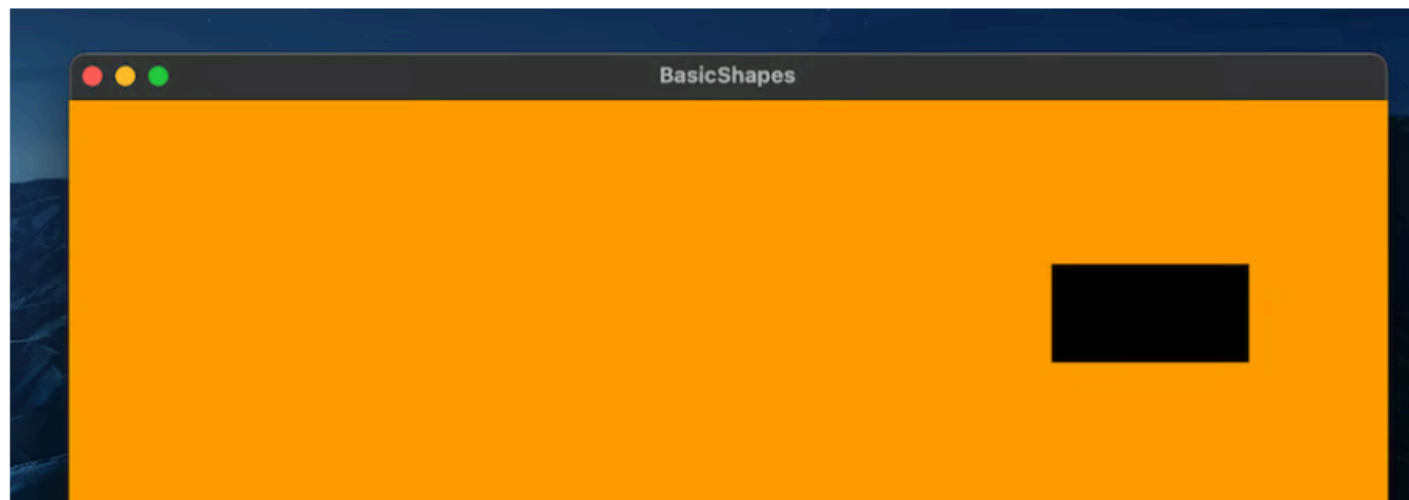
[Check it out](#)

***Editor's note:** This article was last updated on 22 December 2023 to update the list of featured Rust game engines to include Macroquad and honorable mentions such as ggez, Comfy, and rend3.*

The fact that it uses native graphics is quite interesting. It means you can go ahead and run a simple example right now by running the code below:

```
use macroquad::prelude::*;
#[macroquad::main("BasicShapes")]
async fn main() {
    let mut x_position = 0.0;
    loop {
        clear_background(ORANGE);
        draw_rectangle(x_position, 100.0, 120.0, 60.0, BLACK);
        x_position += 2.0;
        if x_position > screen_width() {
            x_position = 0.0;
        }
        next_frame().await
    }
}
```

The example above will display a moving rectangle as shown below:



In the 2D game world (`Node2D` scene tree), no dedicated “rectangle node” exists. You normally do one of these:

- Use `ColorRect` → only works under `Control`, so not for the world.
- Use `Sprite2D` with a plain texture.
- Use `Polygon2D` to define arbitrary shapes, including rectangles.
- Subclass `Node2D` and implement `_draw()` with `draw_rect()`.

For a pure rectangle node in the world, the cleanest is:

gdscript

 Copy  Edit

```
extends Node2D
```

"Bevy is a simple, data-driven game engine."

<https://bevy.org/examples/games/alien-cake-addict/>

```
#[derive(Clone, Copy, PartialEq, Eq, Hash, Debug, Default, States)]
enum GameState {
    #[default]
    Playing,
    GameOver,
}

#[derive(Resource)]
struct BonusSpawnTimer(Timer);

fn main() {
    App::new()
        .add_plugins(DefaultPlugins)
        .init_resource::<<Game>()
        .insert_resource(BonusSpawnTimer(Timer::from_seconds(
            5.0,
            TimerMode::Repeating,
        )))
        .init_state::<<GameState>()
        .enable_state_scoped_entities::<<GameState>()
        .add_systems(Startup, setup_cameras)
        .add_systems(OnEnter(GameState::Playing), setup)
        .add_systems(
            Update,
            (
                move_player,
                focus_camera,
                rotate_bonus,
                scoreboard_system,
                spawn_bonus,
            )
                .run_if(in_state(GameState::Playing)),
        )
        .add_systems(OnEnter(GameState::GameOver), display_score)
        .add_systems(
            Update,
            game_over_keyboard.run_if(in_state(GameState::GameOver)),
        )
        .run();
}
```

Productive Compile Times

Game development is an iterative process. You can't afford to wait for compiles

- With Bevy you can expect 0.8-3.0 seconds with the "fast compiles" configuration
- Compare that to other popular Rust game engines, which can take over 30 seconds to compile a single newline insertion!



Q ✧ I ≡

2

```
4 let questions = vec![
5     ("What does GNU stand for?", "Gnu's Not Unix"),
6     ("Which popular Linux distribution is based on Debian?", "Ubuntu"),
7     ("Who is the creator of the Linux kernel?", "Linus Torvalds"),
8     ("What is the open-source alternative to Microsoft Word?", "LibreOffice"),
9     ("Which version control system is most commonly used?", "Git"),
10 ];
```

```
12 let mut score = 0;
```

```
14     for (question, answer) in questions {
15         println!("{}", question);
16         let mut user_answer = String::new();
17         stdin().read_line(&mut user_answer).unwrap();
18         let user_answer = user_answer.trim().to_lowercase();
19
20         if user_answer == answer {
21             println!("Correct!\n");
22             score += 1;
23         } else {
24             println!("Incorrect. The correct answer is: {}", answer);
25         }
26     }
27 }
```

- Create one at: <https://console.anthropic.com/settings/keys>
- Paste your API key below and hit enter to use the assistant:

You can also assign the `ANTHROPIC_API_KEY` environment variable and restart Zed.

Copilot had issues starting. Please try restarting it. If the issue persists, try reinstalling Copilot.

To use Zed's assistant with Google AI, you need to add an API key. Follow these steps:

- Create one by visiting: <https://aistudio.google.com/app/apikey>
- Paste your API key below and hit enter to use the assistant

You can also assign the `GOOGLE_AI_API_KEY` environment variable and restart Zed.

Get up and running with Llama 3.1, Mistral, Gemma 2, and other large language models with Ollama.

Ollama must be running with at least one model installed to use it in the assistant.

Once installed, try `ollama run llama3.1`

[Download Ollama](#) [All Models](#) Connect

```

}

#[derive(Resource)]
struct BonusSpawnTimer(Timer);

fn main() {
    App::new()
        .add_plugins(DefaultPlugins)
        .init_resource::()
        .insert_resource(BonusSpawnTimer(Timer::from_seconds(
            5.0,
            TimerMode::Repeating,
        )))
        .init_state::()
        .enable_state_scoped_entities::()
        .add_systems(Startup, setup_cameras)
        .add_systems(OnEnter(GameState::Playing), setup)
        .add_systems(
            Update,
            (
                move_player,
                focus_camera,
                rotate_bonus,
                scoreboard_system,
                spawn_bonus,
            )
                .run_if(in_state(GameState::Playing)),
        )
        .add_systems(OnEnter(GameState::GameOver), display_score)
        .add_systems(

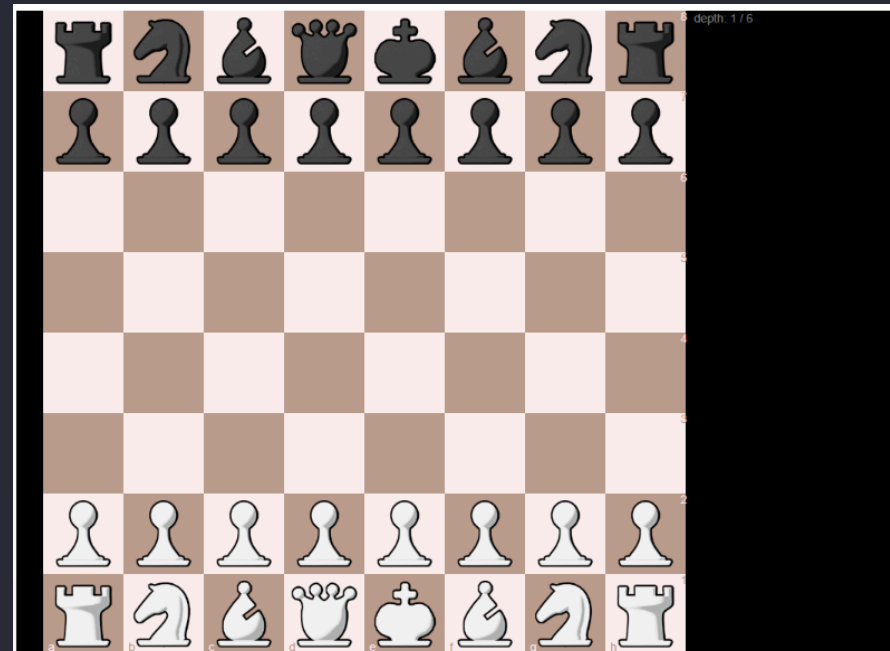
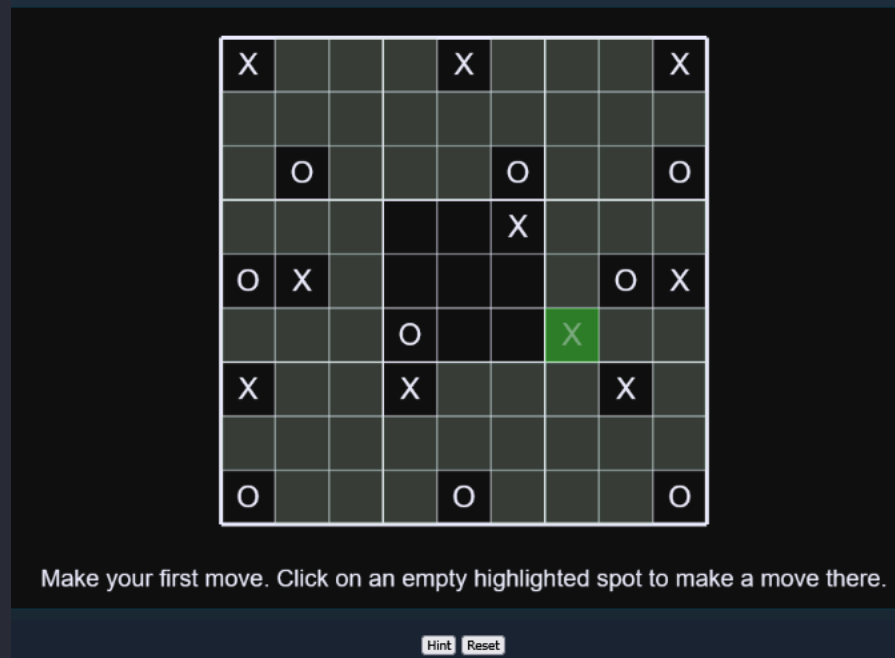
```

Let's use D!

❤️ to Godot and Rust

My D + WebAssembly history

- <https://dkorpel.github.io/tictac/> (2018)
- <https://dkorpel.github.io/ctod/> (2022)
- <https://dkorpel.github.io/dchess> (2023)



dkorpel / Projects / @dkorpel's game engine

Q Type / to search

@dkorpel's game engine

dQw4w9WgXcQ

+ New view

ter by keyword or by field

Discard

Todo2

...

item hasn't been started

Draft

/tile editor

Draft

ari test

In Progress3

...

This is actively being worked on

Draft

Hot reloading / run-time value editing

Draft

Button remapping

Draft

Text automatic word wrap

Getting close1

...

Draft

Text input field

Done19

...

This has been completed

Draft

Platformer physics

Draft

WebGL rendering

Draft

Chrome test

Draft

Collision checking

Draft

Particle system

Draft

OpenGL rendering

Draft

WebAudio

Draft

Viewport / camera

Draft

Pulse audio

Later5

...

Draft

A* Pathfinding

Draft

Debug Console

Draft

UI system

Draft

Rewind / time travel

Draft

Thread safe message passing

15

```
# Run `nix-shell` to use this development environment, 'exit' quit, 'nix-shell --run $SHELL' to update
with import <nixpkgs> {};
mkShell {
  buildInputs = [

    # Dependencies of the executable
    alsa-lib      # Kernel-level audio
    libGL         # Graphics
    pulseaudio    # Desktop manager-level audio
    xorg.libX11   # Window creation, input reading

    # Tools for building
    dmd           # build development builds
    ldc           # build release builds
    esbuild       # to minify javascript
    lld           # for wasm-ld
    wabt          # for wasm-strip
    gnumake       # make
    dub           # dub

    # Running a local HTTP server for WebAssembly
    python3
    pkgs.python3
    pkgs.pythn3Packages.websockets
```

```
# Submitting to itch
butler      # uploading to itch.io

# The better shell (has zsh auto completions!)
zsh

# language servers
serve-d    # D
nil        # nix

# Misc utilities
git
qoi         # convert images to qoi
imagemagick_light # convert images
ffmpeg      # convert audio
lsof        # for testing if the local server is running
coreutils   # for nohup, etc.
];
shellHook = ''
export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:${lib.makeLibraryPath [ libGL pulseaudio xorg.libX11 ]}"
exec zsh    # Automatically switch to zsh
'';
}
```

Tech stack

- object.d: custom WASM runtime (1000 lines, <https://github.com/adamdruppe/webassembly>)
- glfw-d: Windows/X11 window/input
- libsoundio-d: WASAPI / PulseAudio sound
- opengl.d: OpenGL bindings/utils (600 lines)
- webgl.js/gamecanvas.js glue (600 lines)
- collision.d: SAT / ray cast (300 lines)

```
class Room {
    Entity[] entities;
}

class Entity
{
    Vec3 position;

    void create();
    void step(); // fixed time step
    void draw(ref Mesh mesh);
    void drawGui(ref Mesh mesh);
}

class Player : Entity
{
    override step() {...}
}
```

Assets

- qoi.d: image format, .png alternative (300 lines, <https://qoiformat.org/>)
- qoa.d: audio format, .ogg alternative (500 lines, <https://qoaformat.org/>)
- ttf.d: font rendering (1000 lines, <https://github.com/tomolt/libschrift>)

Issues

- code-d / Sublime on NixOS
- Engine developed as we go
- JavaScript `setInterval(..., 1000/60)` = 45 fps on Firefox + power saving
- `requestAnimationFrame(loop)` timer aliasing

```
/// Hack to make i"" strings work, which import core.interpolation which references this symbol
#pragma(mangle, "_D4core13interpolation16__getEmptyStringFNbNbNiNfZAya")
public string __getEmptyString() @nogc pure nothrow @safe => "";
```

Issues



